

7. Demonstrate the usage/sending of a digitally signed document.

A digitally signed document uses cryptographic methods to verify authenticity and integrity.

A. Creating a Digitally Signed Document

1. Prepare the document (e.g., PDF).
2. Open it in a tool supporting digital signatures (Adobe Acrobat, DocuSign, DigiCert).
3. Choose “Sign → Digitally Sign”.
4. Select a digital certificate issued by a trusted authority.
5. Apply the signature and save the document.

B. Sending the Digitally Signed Document

1. Attach the signed PDF to an email or upload it to a secure portal.
2. Recipient opens it using a PDF reader, which validates the signature.

Sample Email Format:

Subject: Submission of Digitally Signed Audit Report

Dear [Recipient],

Please find attached the digitally signed version of the Data Privacy Audit Report.

The digital signature verifies my identity and confirms that the document has not been altered.

Best regards,

[Name]

8. Students need to conduct a data privacy audit of an organization to identify potential vulnerabilities and risks in their data privacy practices.

Organization Selected: Shree Ganesh Retail Pvt. Ltd.

A. Purpose of Audit

To identify risks, vulnerabilities, and compliance gaps under Indian laws such as the IT Act 2000, SPDI Rules (2011), and the Digital Personal Data Protection Act (DPDP Act, 2023).

B. Data Flow Mapping

Data Type	Source	Purpose	Storage	Access	Retention
Name, Phone Number	Loyalty card form	Billing, marketing	Cloud POS (Indian DC)	Store managers, marketing team	5 years
Purchase history	Billing system	Offers, analytics	Local server + cloud backup	Analytics team	7 years
CCTV footage	In-store cameras	Security	Local NVR	Security dept.	30-90 days
UPI/Payment details	QR, POS	Payments	Payment gateway	Finance	As per RBI norms

C. Policy & Procedure Evaluation

Requirement (India)	Observations	Risk
Privacy Notice under DPDP Act	Not displayed in stores; partial online	Medium
Consent for Data Collection	Verbal-only consent	High
Data Storage Security	Outdated local servers	Medium
Data Breach Procedure	No breach notification plan	High
Third-party Agreements	Some vendors lack DPAs	Medium

D. Security Assessment

Strengths:

- Cloud POS uses encryption.
- CCTV access is controlled.

Weaknesses:

- No multi-factor authentication.
- Outdated server patching.
- No anonymization of purchase history.

E. Key Risks Identified

- Non-compliance with consent requirements under DPDP Act.
- Weak access controls on analytics data.
- Vendors not bound by formal data protection agreements.
- Over-retention of customer data.

F. Recommendations

- Introduce standard privacy notices (in-store and online).
- Implement written/informed consent for loyalty data.
- Enable MFA and role-based access.
- Perform regular patching and backups.
- Sign Data Processing Agreements (DPAs) with vendors.
- Reduce retention periods where possible.

9. Students need to explore the requirements of the Data Protection Regulations and develop a plan for ensuring compliance with the regulations.

Applicable Indian Regulations:

- DPDP Act, 2023
- IT Act, 2000 (Section 43A)
- SPDI Rules (2011)
- CERT-In Guidelines (2022)

A. Governance & Roles

- Appoint a Data Protection Officer (recommended).
- Establish a Data Governance Committee for oversight.

B. Consent & Transparency

- Display privacy notices in English, Hindi, and any other language as required.
- Obtain written or digital consent for loyalty program data.
- Notices must include purpose, data categories, grievance officer details, and rights.

C. User Rights Handling

Users may access, correct, delete, or nominate representatives for their data. Organizations should provide an email or self-service portal for requests.

D. Technical Safeguards

- Implement MFA on all internal systems.
- Encrypt databases end-to-end.
- Apply monthly patches.
- Conduct penetration tests via CERT-In empanelled auditors.

E. Data Retention & Disposal

- Contact data: retain 3 years after last use.
- CCTV: retain 30 days.
- Payment data: follow RBI rules.
- Use secure deletion methods.

F. Vendor & Third-Party Compliance

- All vendors must sign Data Processing Agreements (DPAs).
- Vendors must notify breaches within 72 hours.
- Data localization must be followed where required.

G. Breach Response Plan

- Maintain an internal breach register.
- Notify CERT-In within 6 hours of detection (as mandated).
- Inform affected users promptly.

10. Students need to explore ethical considerations in data privacy, such as the balance between privacy and security, the impact of data collection and analysis on marginalized communities, and the role of data ethics in technology development.

A. Balancing Privacy and Security

India's digital systems like Aadhaar, UPI, FASTag, and DigiLocker make services faster, improve identification, and strengthen security. However, they also raise ethical concerns:

- Large-scale data collection can lead to unintended surveillance if not properly regulated.
- People may not always understand or give meaningful consent when their data is used for authentication.
- Security measures must be strong, but not so intrusive that they compromise individuals' right to privacy.

B. Impact on Marginalized Communities

Data systems can unintentionally affect marginalized groups more than others.

- Algorithmic bias may disadvantage rural or minority groups.
- Predictive policing tools may disproportionately target certain communities.
- Micro-lending apps often exploit low-income users through data harvesting.
- Digital illiteracy leads to uninformed consent.

C. Ethics in Technology Development

Developing technology responsibly means considering how data practices affect people's rights.

- Autonomy: Avoid unnecessary permissions and data collection.
- Fairness: Prevent discrimination in AI-based systems.
- Transparency: Explain data use clearly.
- Accountability: Ensure auditability and human oversight in automated decisions.

Conclusion:

Ethical data practices must prioritize fairness, autonomy, and transparency to ensure technology benefits all citizens and avoids reinforcing social inequalities.